

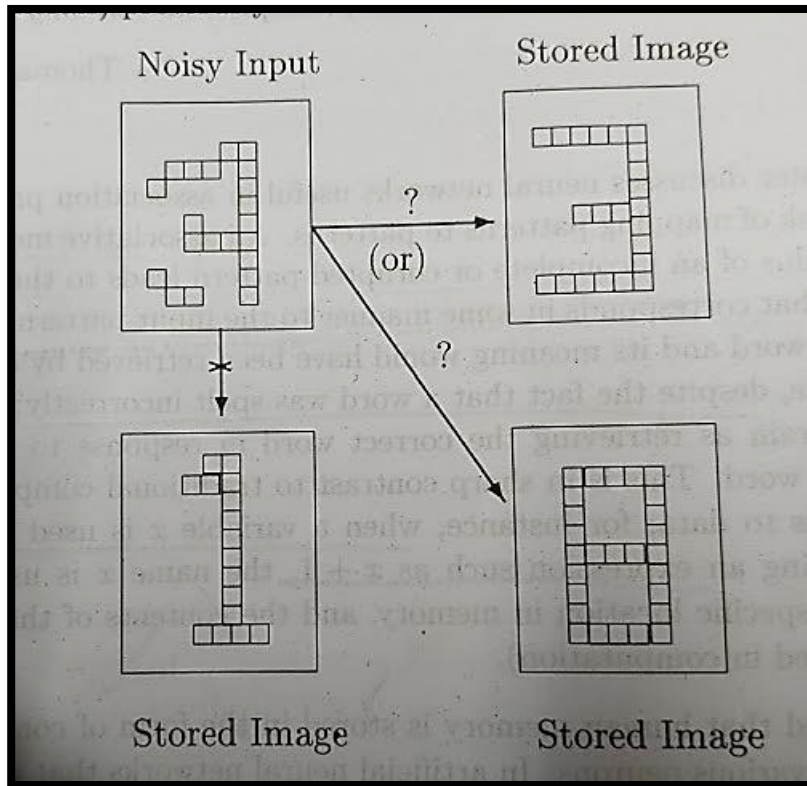
# Associative Learning

Unit -2

# Associative learning

- **Associative learning** in neural networks refers to the process by which **the network forms associations between input patterns and output patterns.**
- This learning process is inspired by how biological systems, like the human brain, learn to associate two or more stimuli or pieces of information.
- **Associative Memory** – is the one in which the stimulus of an incomplete or corrupted pattern leads to the response of a stored pattern, that corresponds in some manner to the input pattern.

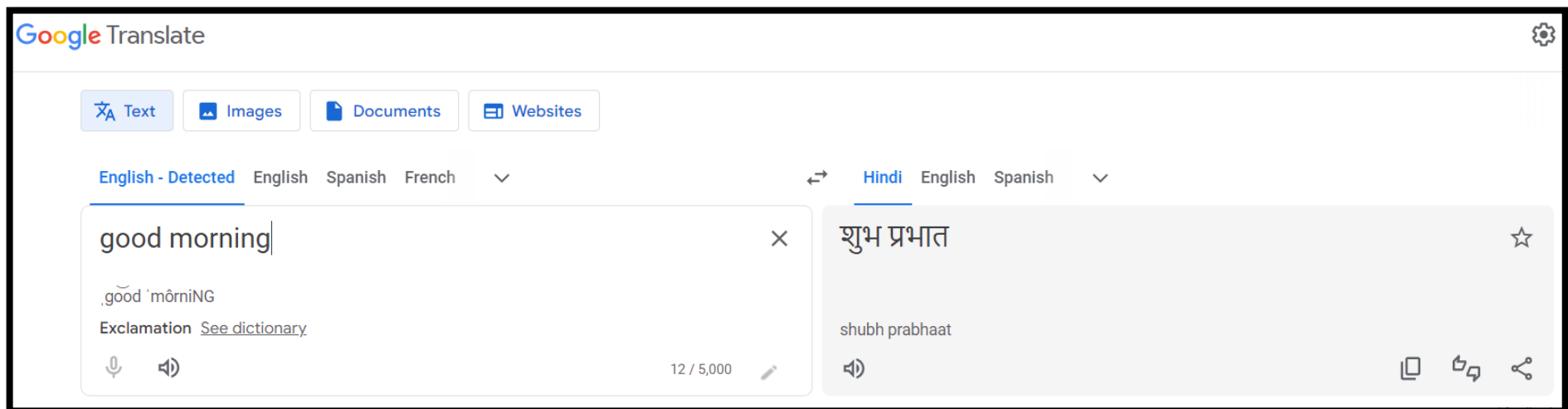
# Example : character recognition



- Character recognition involves auto-association, since output vectors are merely noise-free versions of the input vectors.
- The number of output patterns is called as “memory-size” of auto-associator and each output pattern is called as “memory” or “attractor”.

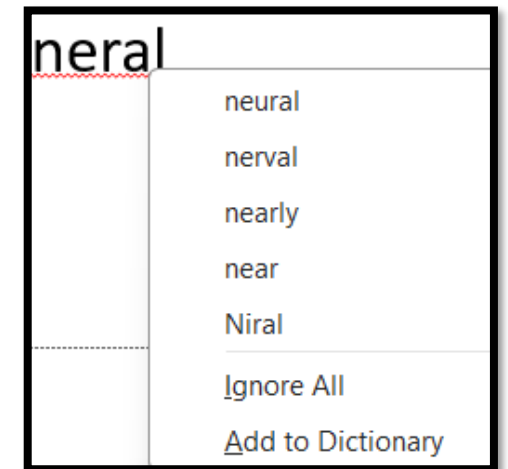
# 2 kinds of association tasks

- A) Hetero-Association
- In **hetero-associative learning**, the neural network learns to associate one pattern with another distinct pattern. This type of learning is called **stimulus-response** learning, where one input pattern triggers a different output pattern.



# 2 kinds of association tasks

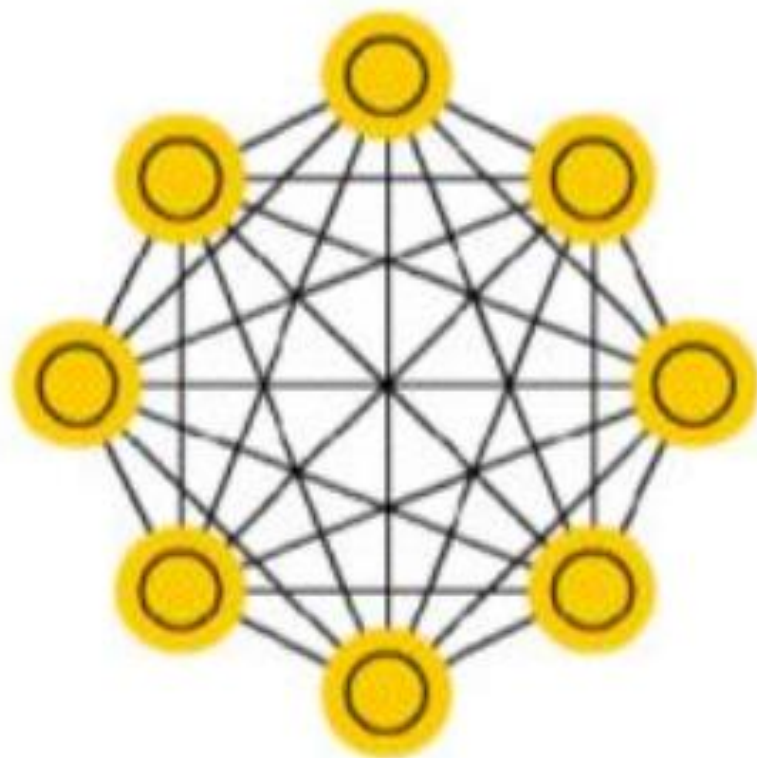
- B) Auto-Association
- In auto-associative learning, a neural network learns to map an input pattern to itself.
- This type of learning is often used in memory recall or pattern completion tasks, where incomplete or noisy inputs are fed into the network, and the network reconstructs the original pattern.
- Both input and output vector range over the same vector space.



# Hopfield Networks

- Introduced by John Hopfield in 1982.
- Used for pattern recognition and storage.
- The network is able to store and recall patterns, even from incomplete or noisy inputs, using the concept of **auto-association**.
- Characteristics of HN:
  - Connections to other neurons → each neuron is connected to all other neurons with a unique strength.
  - Each neuron can have 2 states -1 and +1
  - The network has an associated energy function that decreases as the system moves toward a stable state. This is crucial because Hopfield networks minimize the energy function to retrieve stored patterns.

## Hopfield Network



# Discrete Hopfield Network

- A discrete Hopfield network is a type of recurrent artificial neural network that serves as an associative memory system.
- It is used to store and retrieve patterns (like images, text, or other data) based on the idea that the network will converge to one of the stored patterns after being initialized with an input pattern.
- The network consists of fully connected binary neurons (with states of 1 or -1), and its primary function is to recover stored patterns even from partial or noisy inputs.



# Summary of Steps

- Define patterns to store.
- Calculate the weights between neurons using Hebbian learning.
- Input a noisy or incomplete pattern.
- Iteratively update neuron states using the update rule based on the weighted sum of inputs.
- Minimize the energy of the network with each update.
- Converge to a stable pattern, which is one of the stored patterns or a pattern close to it.

Select a node  $k \in \{1, \dots, n\}$  to be updated;

$$x_{p,\ell}(t+1) = \begin{cases} x_{p,\ell}(t) & \text{if } \ell \neq k \\ \text{sgn} \left( \sum_{j=1}^n w_{\ell,j} x_{p,j}(t) + I_{p,\ell} \right) & \text{if } \ell = k. \end{cases}$$